

Chapter 11: Backtesting and Optimization

The Quest for Reliable Trading Strategies

On August 1, 2012, Knight Capital Group—one of the largest market makers in the U.S.—deployed a new trading algorithm to their production systems. Within the first 45 minutes of trading, the algorithm executed over 4 million trades across 154 stocks, buying high and selling low—the exact opposite of a profitable strategy. By the time they managed to shut down the malfunctioning system, Knight Capital had lost \$440 million, nearly bankrupting the company.

What went wrong? The catastrophic failure wasn't due to a radical new trading idea or complex mathematics. It was much simpler: Knight Capital had failed to properly test their algorithm before deploying it to the live market. An old piece of code that should have been removed was accidentally reactivated, triggering a cascade of erroneous trades.